

Predicting Late Deliveries

Business Problem Statement — TransGlobe Logistics Pvt. Ltd.

Company	TransGlobe Logistics Pvt. Ltd.
Industry	Third-party logistics (3PL) & last-mile delivery
Footprint	25 major Indian cities · 10 carrier partners · 30 warehouses · 3,000+ customers
Volume	22,000+ shipments analyzed

Background

TransGlobe Logistics handles over 22,000 shipments across its network, moving everything from electronics and pharma to apparel and industrial parts. Over the past few quarters, the operations leadership team has noticed a rising number of customer complaints and SLA penalty claims tied to late deliveries — but the company has no systematic way to know **which** shipments are at risk until a customer has already complained.

Currently, delay diagnosis at TransGlobe is entirely reactive: a shipment arrives late, the customer support team logs a complaint, and only then does an ops analyst dig through spreadsheets to figure out whether the carrier, the weather, the warehouse, or the route was to blame. By the time a pattern is found, dozens more shipments have already gone out with the same risk factors.

The Problem

“Predict whether a shipment will be delivered late, and identify the operational factors responsible for delivery delays.”

Late deliveries are costing TransGlobe in three concrete ways:

- **Customer trust** — enterprise clients are threatening to shift volume to competitors after repeated delay incidents
- **SLA penalties** — several enterprise contracts carry financial penalties for late delivery, and TransGlobe is absorbing these costs without any early-warning system
- **Support overhead** — the customer support team spends significant time on reactive delay investigations that could be prevented with advance notice

Why This Matters Now

TransGlobe's leadership has asked the analytics team to move from *reactive* delay reporting to *proactive* risk prediction — flagging high-risk shipments **before dispatch** so operations can intervene: swap to a

more reliable carrier, build in extra buffer time, or proactively notify the customer instead of letting them find out the shipment is late on their own.

Deliverable Requested by Leadership

A binary classification model (**Late_Delivery** = 1/0) built on TransGlobe's shipment, carrier, warehouse, and route data, paired with a dashboard that lets regional ops managers see delay risk by carrier, warehouse, and route in near real time — plus a set of concrete, prioritized recommendations the operations team can act on immediately.

Note: TransGlobe Logistics Pvt. Ltd. is a fictional company created for this portfolio project. The underlying dataset is synthetically generated and does not represent any real organization's data.